Department/Faculty:	Computer Science/ Computing	**Page:**	1 of 7
Course code:	SECR 2043	**Academic Session/Semester:**	2024/2025 – 1
Course name:	OPERATING SYSTEM	**Pre/co requisite:**	SECR1033 COMPUTER ORGANIZATION & ARCHITECTURE
Credit hours:	3		

COURSE OUTLINE

Course synopsis	This course covers introduction to operating systems, which serve as an interface between computer hardware and the user. The operating system is responsible for the management and coordination of processes, sharing of limited resources of the computer. Students will be exposed to the techniques and algorithms that may be applied in designing an operating system. Topics covered include process management, concurrency and synchronization, deadlock, memory management, file management, secondary storage management and I/O management. At the end of the course, the student shall have a clear understanding on the general concepts that underlie of an operating system.
Course coordinator	

Section	**Course Lecturers**	**Office**	**Telephone**	**E-mail**
01. 02	Associate Prof Dr Norafida Ithnin			afida@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO CODE	Weight (%)	*Taxonomies and **generic skills	T&L methods	Assessment methods***
CLO1	***Comprehend*** *the relationship of various operating system mechanisms in handling concurrent processing.*	KW	25	*A2, C2*	Lecture, Active Learning	L1, Q1, Q2, T
CLO2	***Explain*** *how the memory is allocated to processes using different allocation schemes.*	KW	25	*A2, C4*	Lecture, Active Learning	Q3, F
CLO3	***Understand*** *the fundamentals of file management and* ***differentiate*** *file mapping schemes to secondary storage*	KW	25	*A2, C4*	Lecture, Active Learning	L3, F
CLO4	***Ability*** *to program OS related operations or services and deliver ideas effectively to achieve common goals.*	KW	10	C3	Lab	L2, PD
CLO5	***Ability*** *to lead and work effectively in a team to achieve common goals.*	TW	15	*TS1, TS4*	Project-based learning	GR

Refer * Taxonomies of Learning and ** UTM's Graduate Attributes for measurement of outcomes achievement.
*** T – MidTerm Test; Q – Quiz; HW – Homework; L – Lab, GR – Group Project; PD – Project Demo; F – Final Exam etc.

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities.
2.	Project-based learning	Conducted through a given set of case study. Students in a group of 3 (max.) are required to create animation / simulation of algorithms in the case study with relation to the related techniques introduced in the course.

Prepared by:		Certified by:	
Name:	Associate Prof Dr Norafida Ithnin	Name:	Prof. Dr. Md Asri Bin Ngadi (Director)
Signature:		Signature:	
Date:	14 October 2024	Date:	

Department/Faculty:	Computer Science/ Computing	2 of 7
Course code:	SECR 2043	2024/2025 – 1
Course name:	OPERATING SYSTEMS	SECR1033 COMPUTER ORGANIZATION & ARCHITECTURE
Credit hours:	3	

COURSE OUTLINE

Weekly Schedule:

WEEK DATE	TOPICS	*NOTES*
PART 1: OVERVIEW		
Meeting 1 **12 Oct 2024**	**Chapter 1: Introduction** 1. What Operating Systems Do? 2. Computer-System Organization 3. Computer-System Architecture 4. Operating-System Structure 5. Operating-System Operations 6. Process Management 7. Memory Management 8. Storage Management	
	9. Protection and Security 10. Types and Categories of Operating Systems 11. Computing Environments **Chapter 2: Operating-System Structures** 1. Operating-System Services 2. System Calls 3. Type of System Calls 4. System Programs 5. Operating-System Structure 6. System Boot	
PART 2a: PROCESS MANAGEMENT		
Meeting 2 1 Nov 2024	**Chapter 3: Processes** 1. Process Concepts 2. Process Scheduling 3. Operation on Processes 4. Inter-process Communication 5.	**Quiz 1** **(Chap 1)**
	Chapter 4: Thread 1. Overview 2. Multicore Programming	Lab 1: Introduction to linux
Meeting 3 **2 Nov 2024**	**Chapter 5: Process Scheduling** 1. Basic Concepts 2. Scheduling 3. Scheduling Criteria 4. Scheduling Algorithms	**Quiz 2** **(Chap 3)**

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COURSE OUTLINE

PART 2b: PROCESS COORDINATION		
Meeting 4 **22 Nov 2024**	**Chapter 6: Process Synchronization** 1. Background 2. The Critical-Section Problem 3. Peterson's Solution 4. Synchronization Hardware 5. Mutex Locks 6. Semaphores	Lab 2: fork
	Chapter 7: Deadlocks 1. System Model 2. Deadlock Characterization 3. Methods for Handling Deadlocks 4. Deadlock Prevention 5. Deadlock Avoidance 6. Deadlock Detection 7. Recovery from Deadlock	
Week 8	**MID SEMESTER BREAK**	
PART 3: MEMORY MANAGEMENT		
Meeting 5 **23 Nov 2024**	**Chapter 8: Memory-Management Strategies** 1. Background 2. Swapping 3. Memory Allocation	
	4. Segmentation 5. Paging	
Meeting 6 **14 Dec 2024**	**Chapter 9: Virtual-Memory Management** 1. Background 2. Demand Paging 3. Page Replacement 4. Thrashing	**MidTerm Test:** (Chap. 5 & 6)
PART 4: STORAGE MANAGEMENT		
Meeting 7 **3 Jan 2025**	**Chapter 10: File System** 1. File Concepts 2. Directory and Disk Structure 3. Protection	Group Project Commence Lab 3 : Unix file management

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COURSE OUTLINE

	Chapter 11: Implementing File-Systems 1. File-System Structure 2. File-System Implementation 3. Allocation Methods 4. Free-Space Management	**Quiz 3** **(Chap 8)**
Meeting 8 **4 Jan 2025**	**Chapter 12: Mass-Storage Structure** 1. Overview 2. Disk Structure 3. Disk Scheduling 4. Raid Structure	
	REVISION WEEK	

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Thinking skills.

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Teaching and Learning Activities						TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO1	10	1				7.5	18.5
CLO2	10	3	1			25.2	39.2
CLO3	8	2	0.5			22.8	33.3
CLO4	2		1			5	8
CLO5	2		1			5	8
Total SLT	**32**	**6**	**3.5**			**65.5**	**107**

COURSE OUTLINE

Continuous Assessment		PLO	Percentage	Total SLT
1	Lab 1 (CLO1)	PS	5	1.5
2	Lab 2 (CLO4)	PS	5	1.5
3	Lab 3 (CLO3)	PS	5	1.5
4	Quiz 1 (CLO1)	KW	5	0.5
5	Quiz 2 (CLO1)	KW	5	0.5
6	Quiz 3 (CLO2)	KW	5	0.5
7	Midterm Test (CLO1)	KW	20	2
8	Project (Demo – CLO4)	CS2	5	0.5
9	Project (Report – CLO5)	CS1	10	1.5
				10.00
Final Assessment			Percentage	Total SLT
1	Final Examination (CLO2)	KW	10	0.5
2	Final Examination (CLO3)	KW	25	2.5
				3.0
Grand Total SLT				**120**

Learning resources:

Main references/Textbook:
Silbershatz, Galvin, and Gagne, "Operating Systems Concepts, 9th Edition, 2013, John Wiley & Sons.

Additional references:

1. *William Stallings, Operating Systems: Internals and Design Principles, 6th Edition, 2008, Prentice-Hall.*
2. *McHoes, A.M. and Flynn, I.M., Understanding Operating System, 6th Edition, Course Technology, Cengage Learning, 2011.*
3. *H.M. Deitel, Operating Systems, 3rd Edition, Pearson Prentice Hall*
4. *Tanenbaum, Operating System: Design and Implementation, Prentice-Hall.*
5. *D M Dhamdhere, Operating System – A Concept-Based Approach. 2006, Mc Graw-Hill.*

Online:
http://elearning.utm.my

COURSE OUTLINE

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES). Copying of work (texts, lab results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and exams and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Students with less than 80% of total attendance are not allowed to sit for the final exam.
2. Students are required to behave and follow the University's dressing regulations and etiquette all the time.
3. Exercises and tutorials will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exams will not be given, except to students who are sick and submit medical certificates confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of the exam.

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